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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

AD2208-1- SOCIAL WEB ANALYTICS
(For AD)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following is NOT a type of web-based application?
A) Web sites B) Mobile apps
C) Social media platforms D) Operating systems
2. What is the key advantage of using social media for businesses?
A) Reduced operational costs B) Enhanced customer engagement
C) Increased offline sales only D) Avoiding customer feedback
3. Which of the following best describes the impact of social media on business?
A) It only helps with advertisements B) It improves customer engagement, brand awareness, and sales
C) It replaces all traditional marketing efforts D) It is only useful for large companies
4. What is one of the biggest risks businesses face on social media?
A) Increased foot traffic in stores B) Negative customer reviews affecting brand reputation
C) Too much positive engagement D) Lack of customer support features
5. How does social media contribute to a competitive advantage?
A) By limiting customer interaction B) By improving brand visibility and customer trust
C) By reducing the need for analytics D) By decreasing customer engagement
6. What is a key feature of social media listening?
A) Monitoring and analyzing customer sentiments B) Blocking customer feedback
C) Increasing advertising spending D) Avoiding brand interactions
7. What is the primary goal of user experience (UX) design?
A) Increasing website traffic B) Increasing website traffic
C) Creating an intuitive and user-friendly interface D) Creating an intuitive and user-friendly interface
8. What is the primary concern of customer experience (CX) in online platforms?
A) How customers interact with a brand across all touchpoints B) How much money customers spend per visit
C) The number of advertisements shown to users D) The technical complexity of the platform
9. What does "conversion rate" measure in digital marketing?
A) The total revenue from a website B) The number of social media followers
C) The number of people visiting a website D) The percentage of visitors who complete a desired action

10. Which of the following is a key metric in measuring ROI for web marketing?
 A) Number of website visitors B) Number of advertisements posted
 C) Number of competitors D) Revenue generated relative to marketing expenses
11. What is the primary purpose of web analytics?
 A) Collecting random user data B) Measuring and analyzing website performance
 C) Increasing the cost of digital advertising D) Increasing the cost of digital advertising
12. What is one key advantage of using licensed web analytics software over open-source solutions?
 A) It offers advanced technical support and integrations B) It is always free
 C) It is limited in terms of features D) It provides less security
13. What is "Brand Advocacy"?
 A) The legal protection of a brand's logo B) A strategy to reduce the price of products
 C) The process of hiring celebrities for endorsements D) When satisfied customers or employees voluntarily promote a brand
14. What metric in web analytics measures the percentage of visitors who leave a website after viewing only one page?
 A) Click-through Rate (CTR) B) Bounce Rate
 C) Exit Rate D) Conversion Rate
15. Which of the following is NOT tracked in web analytics?
 A) The thoughts of the user B) Mouse movement on a page
 C) Number of clicks D) User location
16. What is the primary function of web analytics?
 A) Increasing web development costs B) Replacing traditional sales methods
 C) Reducing online advertising expenses D) Tracking and analyzing website traffic and user behavior
17. What is a major drawback of keyword-based search in text mining?
 A) It may return irrelevant results if context is not considered B) It only works with structured data
 C) It always provides perfect results D) It is not used in social media mining
18. How can answer ranking improve search results in text mining?
 A) By prioritizing results based on relevance to the query B) By displaying all results randomly
 C) By limiting the number of search results D) By ignoring frequently used words
19. Which of the following is an example of a classification algorithm?
 A) Decision Trees B) PageRank
 C) K-Means Clustering D) Principal Component Analysis
20. What type of data is commonly used in keyword search over XML?
 A) Structured hierarchical data B) Encrypted binary data
 C) Only numerical data D) Only unstructured text data

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is the significance of user experience and customer experience in web analytics? How do they differ?	6	L*2	1	1
	b) Explain the role of Return on Investment (ROI) in evaluating the success of social media marketing strategies	6	2	1	1
	c) Describe the technical requirements for implementing web analytics in an organization.	4	2	1	1
2.	a) Explain the importance of understanding customer sentiments in social media analytics. Provide an example of how customer sentiment can influence business decisions.	6	2	1	1

- b) Compare the advantages of using R for data analysis over traditional spreadsheet-based approaches. 6 2 2 1
- c) What are the key factors to consider when participating in a people-centric approach to data collection? 4 2 2 1
3. a) What are the main challenges in analyzing big data, and how can linked data help address these challenges? 8 2 1 1
- b) Write a short R script to read a CSV file containing website traffic data and calculate basic descriptive statistics (e.g., mean, median, mode). 8 3 2 1

Unit – II

4. a) Explain KPIs in the context of social analytics. 4 2 3 1
- b) A company wants to improve its online sales strategy and is analyzing its website's performance. Identify and classify key user actions on the website as macro or micro conversions. Justify your classification with relevant examples. 8 3 3 1
- c) Explain the significance of PULSE metrics in addressing business and technical issues. 4 2 4 1
5. a) What are the differences between on-site and off-site web analytics? 4 2 3 1
- b) You are a web analyst working for an online education platform. The platform has three key pages: Landing Page, Course Page, and Payment Page. The following data has been collected for a specific time period:

Page	Total Visits	Bounces	Exits	Conversions	Engaged Visits
Landing Page	15,000	4,500	6,000	1,200	9,000
Course Page	10,000	2,000	3,500	1,500	7,000
Payment Page	5,000	500	1,500	3,000	4,000

Calculate the Bounce, Exit, Conversion and Engagement rates also
Calculate the overall website performance metrics.

- c) Explain the concept of bounce rate and exit rate in web analytics. 8 3 4 1
6. a) Explain the HEART metrics framework in detail. How can it be applied to analyze user behavior on a social media platform? 4 2 4 1
- b) What are the critical differences between standard metrics and critical metrics in social analytics? Provide examples. 8 2 3 1
- 8 3 4 1

Unit – III

7. a) Explain data mining in social media and explain its key motivations. How does data representation play a role in this process? 8 2 5 1
- b) Explain the process of data mining in social media, highlighting at least two methods commonly used for mining data from social networking sites. Provide illustrative examples. 8 2 5 1
8. a) Explain the concept of text mining in social networks. Discuss how keyword search is implemented in relational and XML data. 8 2 5 1
- b) Compare classification algorithms and clustering algorithms in the context of social network analysis. Provide examples where each would be suitable. 8 2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
